

The Auditable Landauer Principle (ALP)

A structured, testable framework for identity-aware thermodynamics

Executive Summary

ALP extends Landauer’s principle to **structured information loss**. We formalize two distinct kinds of “forgetting” that occur in real systems:

1. **Haecceity** (which-token) loss via a quotient by permutations; and
2. **Multiplicity/Aggregation** (how-many/amount) loss via coarse-graining (summing aspects).

For each, we supply a **reversible audit lift** that preserves the otherwise-lost information as an explicit record. The resulting **Auditable Landauer bound** states that the *irreversible* erasure of audit information carries a heat cost proportional to the conditional entropies of those variables given the retained side information.

ALP (core claim)

The minimal average heat for erasing identity information is

$$\langle Q \rangle_{\text{erase}} \geq kT \ln 2 [H(L | S) + H(M | S)]$$

where **L** are labels (haecceity), **M** is the multiset of aggregated aspects (multiplicity), and **S** denotes *all* retained, accessible side information (including any deliberate **audit**). If a complete audit of a component is preserved, its conditional entropy vanishes and that term drops from the fundamental cost. The bill is paid precisely when/where the audit is irreversibly reset.

Operational Haecceity (no metaphysics)

Definition. In a given experimental design with observables **O**, we say haecceity is *operationally carried* by an audit variable **S** if it renders labels irrelevant for predictions:

$$I(L; O | S) = 0.$$

ALP is purely about **information flow and control**, not about intrinsic “this-ness.”

Two-Stage Forgetting as Maps

- **Tokens:** $X \in T^N$
- **Haecceity quotient:** $q : T^N \rightarrow T^N / S_N = \text{Sym}^N(T)$
- **Aggregation of an aspect** $\phi : T \rightarrow A$ (monoid $(A, +)$):
 $\mu : \text{MSet}(A) \rightarrow A, \mu(\{a_i\}) = \sum_i a_i$

Audit lifts (reversible variants)

- **Haecceity:**

$$\hat{q}(x) = (q(x), \pi(x)) \in \text{Sym}^N(T) \times S_N,$$

where π is the permutation mapping a canonical representative back to x .

- **Aggregation:**

$$\hat{\mu}(Y) = (\sum a_i, \text{bag}\{a_i\}) \in A \times \text{MSet}(A),$$

i.e., keep the multiset ("bag") of aspects alongside the sum.

Both lifts are **injective** on realized trajectories; dropping the audit components produces the usual (lossy) maps.

The Auditable Landauer Bound

With identity variables **L** (labels) and **M** (multiset), and retained side data **S** (including audits), erasures satisfy: $\langle Q \rangle \geq kT \ln 2 [H(L | S) + H(M | S)]$.

- Handles **non-uniform priors** and **hidden records** via conditional entropy.
 - Collapses to $\langle Q \rangle \geq kT \ln(N!)$ only in the special case of uniform labels and no side info.
 - Reflects the principle: **hidden** $\neq$ **erased** (information in S softens the floor).
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Validation Ladder (clean pass/fail)

C) Quantum audit toggle (HOM)

Goal: Validate the *logic* of audit on/off for haecceity (no thermodynamics).

Method: Two-photon interference with which-path markers (audit) added/erased.

- Visibility $V = [C(\infty) - C(0)]/C(\infty)$.
- **Pass:** marker **on** $\Rightarrow V \leq V_{\text{on,max}}$; no marker $\Rightarrow V \geq V_{\text{off,min}}$; **erased** $\Rightarrow V$ statistically indistinguishable from no-marker.

A) Landauer test (optical tweezer, M-state)

Goal: Validate the *thermodynamic* inequality with explicit audits.

Method: Encode $L \in \{1..M\}$ in multi-well potential; quasi-static erase; compare heat to audit-aware bound.

- **Pass:** $\langle Q \rangle \geq kT \ln 2 H(L | S)$ (one-sided test).
- **Adiabaticity:** Excess $\Delta Q(\tau) = Q - kT \ln 2 H(L | S)$ decreases with protocol time; last-quartile mean within tolerance.
- **Swap vs erase:** With reversible swap (audit retained), the $kT \ln 2 H$ floor attributable to L disappears; it reappears when the audit is reset.

B) Isotope audit (aggregation logic)

Goal: Validate the *aggregation* story: type-level invariants vs token history.

Method: Mix two isotopically distinct samples; assay composition.

- **Pass:** Mixture χ^2 consistent with mass-weighted average; token-level mutual information ≈ 0 absent explicit tags.
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Engineering Note: Cost of Maintaining the Audit

ALP distinguishes **logical erasure** (bounded by the inequality) from **maintenance overhead** (control, isolation, error correction). Keeping an audit defers the Landauer bill until the audit is eventually reset; maintenance costs depend on device physics and do not affect the *fundamental* bound unless erasure occurs.

Analysis & Tools (ready to use)

A lightweight analysis kit (Python) accompanies this framework:

- `landauer_tweezer_analysis.py` — heat vs ALP bound with bootstrap pass/fail.
 - `landauer_excess_check.py` — adiabaticity: ΔQ trend + asymptote test.
 - `landauer_power.py` — sample-size (cycles) vs detectable margin.
 - `hom_visibility_analysis.py` — HOM visibility statistics.
 - `isotope_audit_analysis.py` — mixture χ^2 and token-information estimate.
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External References (selected)

- **Landauer’s original principle:**
R. Landauer, “Irreversibility and Heat Generation in the Computing Process,” *IBM Journal of Research and Development* **5**(3), 183–191 (1961). doi:10.1147/rd.53.0183
- **Logical reversibility:**
C. H. Bennett, “Logical Reversibility of Computation,” *IBM Journal of Research and Development* **17**(6), 525–532 (1973). doi:10.1147/rd.176.0525
- **Experimental saturation of $kT \ln 2$ (optical trap):**
A. Bérut, A. Arakelyan, A. Petrosyan, S. Ciliberto, R. Dillenschneider, E. Lutz, “Experimental verification of Landauer’s principle,” *Nature* **483**, 187–189 (2012). doi:10.1038/nature10872
- **Finite-size corrections / equality form:**
D. Reeb, M. M. Wolf, “An improved Landauer principle with finite-size corrections,” *New Journal of Physics* **16**, 103011 (2014). doi:10.1088/1367-2630/16/10/103011
- **Accessible HOM audit toggle (teaching/bench):**
N. S. DiBrita, E. J. Galvez, “An easier-to-align Hong–Ou–Mandel interference demonstration,” *American Journal of Physics* **91**(4), 307–315 (2023). doi:10.1119/5.0119906

- **Original HOM effect:**

C. K. Hong, Z. Y. Ou, L. Mandel, "Measurement of subpicosecond time intervals between two photons by interference," *Physical Review Letters* **59**, 2044–2046 (1987).

These references align with ALP's logical (audit), thermodynamic (Landauer), and experimental (HOM, optical tweezer) pillars.

One-Paragraph Abstract (for papers/proposals)

We extend Landauer's principle to structured information loss by introducing the **Auditable Landauer Principle (ALP)**. Identity is treated operationally: haecceity is the information needed to distinguish otherwise-similar tokens via an externally accessible audit variable S . We isolate two non-invertibilities—(i) a haecceity quotient q (label/orbit forgetting) and (ii) multiplicity aggregation μ (scale/amount forgetting)—and provide audit lifts that render each reversible: $(q(x), \pi(x))$ for haecceity and $(\sum a_i, \text{bag}\{a_i\})$ for aggregation. We prove a conditional-entropy bound that handles hidden records: $\langle Q \rangle \geq kT \ln 2 [H(L | S) + H(M | S)]$. A validation ladder demonstrates logic (Hong–Ou–Mandel audit toggle), thermodynamics (optical-tweezer Landauer for M -state memories with adiabatic approach), and macroscopic intuition (isotope mixing), each with explicit pass/fail inequalities and analysis code. ALP clarifies that audit retention defers—rather than violates—Landauer's cost, enabling temporal and architectural control over dissipation in information-processing systems.
